WORKED SOLUTION

STEP III 2005, Question 2

Pure • Differential equations

STEP III 2005, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s3-q2

The key idea

The differential equation is separable, and the integration gives logarithms that combine into the stated form. For the distance question, minimise the *square* of the distance, $D = x^2 + y^2$, which avoids square roots. On the curve y^2 is an explicit function of x, so D becomes a function of x alone, and its derivative factorises neatly. Which stationary points appear depends on how c compares with a^2.

Solution

The general solution

Separating the variables (for $y \neq 0$),

$$\int \frac{1}{y}\,\mathrm{d}y = -\int \frac{x}{x^2+a^2}\,\mathrm{d}x \quad \Longrightarrow \quad \ln|y| = -\tfrac{1}{2}\ln(x^2+a^2) + k,$$

where k is a constant. Note that $x^2 + a^2 > 0$ for every x because $a \neq 0$, so the logarithm on the right is defined everywhere. Doubling and taking exponentials,

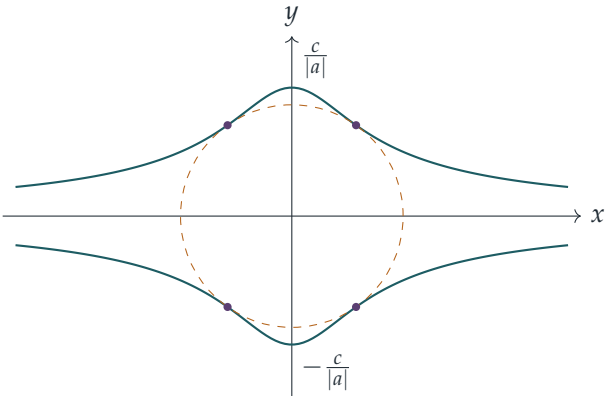
$$y^2 = \frac{\mathrm{e}^{2k}}{x^2+a^2}, \qquad \text{so} \qquad y^2(x^2+a^2) = c^2,$$

with $c^2 = \mathrm{e}^{2k}$. The solution $y = 0$, which we lost when dividing by y, is the case $c = 0$, so c is indeed an arbitrary constant. Explicitly, the general solution is $y = c/\sqrt{x^2+a^2}$, where c may have either sign. Since only c^2 appears in the curve's equation, we take $c > 0$ from now on.

Sketching the curve

The curve $y^2(x^2+a^2) = c^2$ consists of the two branches $y = \pm c/\sqrt{x^2+a^2}$. It is symmetric in both axes (only x^2 and y^2 appear) and never meets the x-axis. On the upper branch, y is greatest when $x^2 + a^2$ is least, giving the maximum $c/|a|$ at $x = 0$.

As $|x| \to \infty$ we have $y \approx \pm c/|x| \to 0$, so the x-axis is an asymptote. The figure shows the case $a = 1$, $c = 2$, together with the circle that we'll find in part (ii).



The derivatives of $x^2 + y^2$

Let $D = x^2 + y^2$. Using the differential equation and then $y^2 = c^2/(x^2 + a^2)$,

$$\frac{\mathrm{d}D}{\mathrm{d}x} = 2x + 2y\frac{\mathrm{d}y}{\mathrm{d}x} = 2x - \frac{2xy^2}{x^2 + a^2} = 2x\left(1 - \frac{c^2}{(x^2 + a^2)^2}\right).$$

This holds on both branches, since it involves only y^2. Differentiating again with the product rule, and using $\frac{\mathrm{d}}{\mathrm{d}x}(x^2 + a^2)^{-2} = -4x(x^2 + a^2)^{-3}$,

$$\frac{\mathrm{d}^2D}{\mathrm{d}x^2} = 2\left(1 - \frac{c^2}{(x^2 + a^2)^2}\right) + 2x \cdot \frac{4c^2x}{(x^2 + a^2)^3} = 2\left(1 - \frac{c^2}{(x^2 + a^2)^2}\right) + \frac{8c^2x^2}{(x^2 + a^2)^3},$$

as required.

The stationary points of D are where $x = 0$ or $(x^2 + a^2)^2 = c^2$. Since $x^2 + a^2 > 0$ and $c > 0$, the second condition means $x^2 + a^2 = c$, that is, $x^2 = c - a^2$.

Part (i)

Suppose $0 < c < a^2$. For every x we have $x^2 + a^2 \geq a^2 > c$, so $(x^2 + a^2)^2 > c^2$ and the bracket in $\mathrm{d}D/\mathrm{d}x$ is strictly positive. Hence $\mathrm{d}D/\mathrm{d}x$ has the same sign as x: D decreases for $x < 0$ and increases for $x > 0$. So D has a *global* minimum at $x = 0$ and nowhere else. (Consistently, the second derivative there is $2(1 - c^2/a^4) > 0$.)

At $x = 0$, $y^2 = c^2/a^2$, so the nearest points are

$$\left(0,\ \pm\frac{c}{a}\right),$$

at distance $c/|a|$ from the origin. This is the required result.

Part (ii)

Now suppose $c > a^2$. There are three stationary points: $x = 0$ and $x = \pm\sqrt{c - a^2}$.

Ask yourself Is $x = 0$ still the nearest point?

No. At $x = 0$ the second derivative is $2(1 - c^2/a^4)$, which is negative because $c > a^2 > 0$ gives $c^2 > a^4$. So $x = 0$ is now a local *maximum* of D. At $x^2 = c - a^2$ we have $x^2 + a^2 = c$, so the bracket vanishes and

$$\frac{\mathrm{d}^2 D}{\mathrm{d}x^2} = \frac{8c^2(c - a^2)}{c^3} = \frac{8(c - a^2)}{c} > 0,$$

giving local minima. To see that they are global, look at the sign of $\mathrm{d}D/\mathrm{d}x$. For $x > 0$ the bracket is negative when $x^2 < c - a^2$ and positive when $x^2 > c - a^2$, so on $x \geq 0$ the function D decreases to $x = \sqrt{c - a^2}$ and increases after it. Hence D is least on $x \geq 0$ at $x = \sqrt{c - a^2}$, and by symmetry it is least on $x \leq 0$ at $x = -\sqrt{c - a^2}$.

At these points $y^2 = c^2/(x^2 + a^2) = c^2/c = c$. So the points on the curve nearest the origin are the four points

$$\left(\pm\sqrt{c - a^2},\ \pm\sqrt{c}\right)$$

(all four sign combinations), at distance $\sqrt{2c - a^2}$ from the origin.

As a cross-check, the distance squared at $x = 0$ is c^2/a^2, and

$$\frac{c^2}{a^2} - (2c - a^2) = \frac{(c - a^2)^2}{a^2} > 0,$$

so the points on the y-axis are indeed farther away.

Checks

- Substituting $y = c/\sqrt{x^2 + a^2}$ back into the differential equation works: $y' = -cx(x^2 + a^2)^{-3/2} = -xy/(x^2 + a^2)$.
- With $a = 1$, $c = 2$, a brute-force search of $x^2 + 4/(x^2 + 1)$ finds the least value $3 = 2c - a^2$ at $x = \pm 1 = \pm\sqrt{c - a^2}$, with $y^2 = 2 = c$. These are the four dots on the figure, where the dashed circle of radius $\sqrt{3}$ touches the curve.
- At $c = a^2$ the two answers agree: $\sqrt{c - a^2} = 0$ and $\sqrt{c} = c/|a| = |a|$, so the four points merge into $(0, \pm|a|)$.

Common slips

- Concluding from "$\mathrm{d}D/\mathrm{d}x = 0$ and $\mathrm{d}^2D/\mathrm{d}x^2 > 0$" that a point is the *least* distance. That only gives a local minimum; the sign of $\mathrm{d}D/\mathrm{d}x$ settles the global question.
- Forgetting the lower branch, and so giving only two points in part (ii) instead of four.
- Writing $x^2 + a^2 = \pm c$: the negative option is impossible, since $x^2 + a^2 > 0$.

Beyond STEP

Minimising $x^2 + y^2$ subject to $y^2(x^2 + a^2) = c^2$ is a constrained optimisation problem. At university you'd meet Lagrange multipliers, which handle such problems without solving for one variable: at the nearest point, the curve's normal points along the radius, which is why the dashed circle touches the curve. The way the single minimum at $x = 0$ splits into two as c passes a^2 is a simple example of a *pitchfork bifurcation*. For more on separable equations and their solution curves, see Boyce and DiPrima, *Elementary Differential Equations*.

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